

Signaling through ERK1/2 controls myelin thickness during myelin repair in the adult central nervous system.

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Oligodendrocytes, the myelin-forming cells of the CNS, exquisitely tailor the thickness of individual myelin sheaths to the diameter of their target axons to maximize the speed of action potential propagation, thus ensuring proper neuronal connectivity and function. Following demyelinating injuries to the adult CNS, newly formed oligodendrocytes frequently generate new myelin sheaths. Following episodes of demyelination such as those that occur in patients with multiple sclerosis, however, the matching of myelin thickness to axon diameter fails leaving remyelinated axons with thin myelin sheaths potentially compromising function and leaving axons vulnerable to damage. How oligodendrocytes determine the appropriate thickness of myelin for an axon of defined size during repair is unknown and identifying the signals that regulate myelin thickness has obvious therapeutic implications. Here, we show that sustained activation of extracellular-regulated kinases 1 and 2 (ERK1/2) in oligodendrocyte lineage cells results in accelerated myelin repair after injury, and is sufficient for the generation of thick myelin sheaths around remyelinated axons in the adult mouse spinal cord. Our findings suggest a model where ERK1/2 MAP kinase signaling acts as a myelin thickness rheostat that instructs oligodendrocytes to generate axon-appropriate quantities of myelin.

R-Ras1 and R-Ras2 regulate mature oligodendrocyte subpopulations.

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In the mammalian central nervous system, axonal myelination, executed by mature oligodendrocytes (MOLs), enables rapid neural transmission. Conversely, myelin deficiencies are hallmark features of multiple sclerosis, optic neuromyelitis, and some leukodystrophies. Recent studies have highlighted that MOLs are heterogeneous; however, how MOL subpopulations are specified and balanced in physiological settings is poorly understood. Previous works have demonstrated an essential role of the small GTPases R-Ras1 and R-Ras2 in the survival and myelination of oligodendrocytes. In this study, we aimed to determine how R-Ras1 and R-Ras2 contribute to the heterogeneity of MOL subpopulations. Our results evidence that R-Ras1 and R-Ras2 affect specification into the distinct subpopulations MOL1, MOL2, and MOL5/6, which in turn vary in their dependence of these GTPases. In R-Ras1 and/or R-Ras2 mutant mice, we observed an increase in the MOL1 subpopulation and a decrease in the MOL2 and MOL5/6 subpopulations. We identified R-Ras1 and R-Ras2 as key elements in balancing the heterogeneity of MOLs. Our results contribute to the understanding of the molecular mechanisms underlying the heterogeneity of MOLs and the myelination processes, which is crucial for innovating regenerative therapies for nervous system disorders.

A new player in the game: identification of C1ql1 as a novel factor driving OPC differentiation.

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Oligodendrocytes (OLGs) are the myelin-producing cells in the central nervous system (CNS). Following injury, these cells are prone to death, leading to demyelination and, eventually, axonal loss and

neurodegeneration. Upon injury, the damaged CNS repopulates the lesion with oligodendrocyte precursor cells (OPCs) that consequently mature into OLGs to repair the myelin damage and prevent further axonal loss. In this issue, Altunay et al. identified that complement component 1, q subcomponent-like-1 (C1ql1), a factor known to play a role in neuron-neuron synapses, is also expressed by OPCs and drives their differentiation into OLGs. These data suggest that C1ql1 or other downstream factors could be therapeutic targets in the context of demyelinating disorders in which remyelination fails, such as in multiple sclerosis (MS).

Pharmacogenomic screening identifies and repurposes leucovorin and dyclonine as pro-oligodendrogenic compounds in brain repair.

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Oligodendrocytes are critical for CNS myelin formation and are involved in preterm-birth brain injury (PBI) and multiple sclerosis (MS), both of which lack effective treatments. We present a pharmacogenomic approach that identifies compounds with potent pro-oligodendrogenic activity, selected through a scoring strategy (OligoScore) based on their modulation of oligodendrogenic and (re)myelination-related transcriptional programs. Through in vitro neural and oligodendrocyte progenitor cell (OPC) cultures, ex vivo cerebellar explants, and in vivo mouse models of PBI and MS, we identify FDA-approved leucovorin and dyclonine as promising candidates. In a neonatal chronic hypoxia mouse model mimicking PBI, both compounds promote neural progenitor cell proliferation and oligodendroglial fate acquisition, with leucovorin further enhancing differentiation. In an adult MS model of focal de/remyelination, they improve lesion repair by promoting OPC differentiation while preserving the OPC pool. Additionally, they shift microglia from a pro-inflammatory to a pro-regenerative profile and enhance myelin debris clearance. These findings support the repurposing of leucovorin and dyclonine for clinical trials targeting myelin disorders, offering potential therapeutic avenues for PBI and MS.

Nonvesicular lipid transfer drives myelin growth in the central nervous system.

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Oligodendrocytes extend numerous cellular processes that wrap multiple times around axons to generate lipid-rich myelin sheaths. Myelin biogenesis requires an enormously productive biosynthetic machinery for generating and delivering these large amounts of newly synthesized lipids. Yet, a complete understanding of this process remains elusive. Utilizing volume electron microscopy, we demonstrate that the oligodendroglial endoplasmic reticulum (ER) is enriched in developing myelin, extending into and making contact with the innermost myelin layer where growth occurs. We explore the possibility of transfer of lipids from the ER to myelin, and find that the glycolipid transfer protein (GLTP), implicated in nonvesicular lipid transport, is highly enriched in the growing myelin sheath. Mice with a specific knockout of Gltp in oligodendrocytes exhibit ER pathology, hypomyelination and a decrease in myelin glycolipid content. In summary, our results demonstrate a role for nonvesicular lipid transport in CNS myelin growth, revealing a cellular pathway in developmental myelination.

Decreased Regenerative Capacity of Oligodendrocyte Progenitor Cells (NG2-Glia) in the Ageing Brain: A Vicious Cycle of Synaptic Dysfunction, Myelin Loss and Neuronal Disruption?

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Oligodendrocytes are specialised glial cells that myelinate CNS axons. Myelinated axons are bundled together into white matter tracts that interconnect grey matter areas of the brain and are essential for rapid, integrated neuronal communication and cognitive function. Life-long generation of oligodendrocytes is required for myelination of new neuronal connections and repair of myelin lost through natural 'wear and tear'. This is the function of a substantial population of adult oligodendrocyte progenitors (OPs). Notably, there is white matter shrinkage and decreased myelination in the ageing brain, which is accelerated in dementia. The underlying causes of myelin loss in dementia are unresolved, but it implies a decline in the regenerative capacity of OPs. A feature of OPs is that they form neuron-glial synapses and respond to glutamate released by neurons via a range of glutamate receptors. Glutamate neurotransmission onto OPs is proposed to regulate their proliferation and differentiation into myelinating oligodendrocytes. Here, we discuss evidence that deregulation of glutamate neurotransmission in dementia and compromised generation of oligodendrocytes from OPs are key features of myelin loss and associated cognitive decline.

A primary culture method for the easy, efficient, and effective acquisition of oligodendrocyte lineage cells from neonatal rodent brains.

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Oligodendrocytes (OL) are myelin-forming glial cells in the central nervous system. In vitro primary OL culture models offer the benefit of a more readily controlled environment that facilitates the examination of diverse OL stages and their intricate dynamics. Although conventional methods for primary OL culture exist, their performance in terms of simplicity and efficiency can be improved. Here, we introduce a novel method for primary OL culture, namely the E3 (easy, efficient, and effective) method, which greatly improves the simplicity and efficiency of the primary OL culture procedure using neonatal rodent brains. We also provided the optimal media composition for the augmentation of oligodendrocyte progenitor cell (OPC) proliferation and more robust maturation into myelin-forming OLs. Overall, E3 offers an undemanding method for obtaining primary OLs with high yield and quality. Alongside its value as a practical tool, in vitro characteristics of the OL lineage additionally identified during the development of the E3 method have implications for advancing research on OL physiology and pathophysiology.

Temporal oligodendrocyte lineage progression: in vitro models of proliferation, differentiation and myelination.

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Oligodendrocytes are neuroglial cells responsible, within the central nervous system, for myelin sheath formation that provides an electric insulation of axons and accelerate the transmission of electrical signals. In order to be able to produce myelin, oligodendrocytes progress through a series of

differentiation steps from oligodendrocyte precursor cells to mature oligodendrocytes (migration, increase in morphologic complexity and expression pattern of specific markers), which are modulated by cross talk with other nerve cells. If during the developmental stage any of these mechanisms is affected by toxic or external stimuli it may result into impaired myelination leading to neurological deficits. Such being the case, several approaches have been developed to evaluate how oligodendrocyte development and myelination may be impaired. The present review aims to summarize changes that oligodendrocytes suffer from precursor cells to mature ones, and to describe and discuss the different in vitro models used to evaluate not only oligodendrocyte development (proliferation, migration, differentiation and ability to myelinate), but also their interaction with neurons and other glial cells. First we discuss the temporal oligodendrocyte lineage progression, highlighting the differences between human and rodent, usually used as tissue supply for in vitro cultures. Second we describe how to perform and characterize the different in vitro cultures, as well as the methodologies to evaluate oligodendrocyte functionality in each culture system, discussing their advantages and disadvantages. Finally, we briefly discuss the current status of in vivo models for oligodendrocyte development and myelination.

A fatal alliance: Glial connexins, myelin pathology and mental disorders.

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Mature oligodendrocytes are myelin forming glial cells which are responsible for myelination of neuronal axons in the white matter of the central nervous system. Myelin pathology is a major feature of severe neurological disorders. Oligodendrocyte-specific gene mutations and/or white matter alterations have also been addressed in a variety of mental disorders. Breakdown of myelin integrity and demyelination is associated with severe symptoms, including impairments in motor coordination, breathing, dysarthria, perception (vision and hearing), and cognition. Furthermore, there is evidence indicating that myelin sheath defects and white matter pathology contributes to the affective and cognitive symptoms of patients with mental disorders. Oligodendrocytes express the connexins GJC2; mCx47 [human (GJC2) and mouse (mCx47) connexin gene nomenclature according to Söhl and Willecke (2003)], GJB1; mCx32, and GJD1; mCx29 in both white and gray matter. Preclinical findings indicate that alterations in connexin expression in oligodendrocytes and astrocytes can induce myelin defects. GJC2; mCx47 is expressed at early embryonic stages in oligodendrocyte precursors cells which precedes central nervous system myelination. In adult humans and animals GJC2, respectively mCx47 expression is essential for oligodendrocyte function and ensures adequate myelination as well as myelin maintenance in the central nervous system. In the past decade, evidence has accumulated suggesting that mental disorders can be accompanied by changes in connexin expression, myelin sheath defects and corresponding white matter alterations. This dual pathology could compromise inter-neuronal information transfer, processing and communication and eventually contribute to behavioral, sensory-motor, affective and cognitive symptoms in patients with mental disorders. The induction of myelin repair and remyelination in the central nervous system of patients with mental disorders could help to restore normal neuronal information propagation and ameliorate behavioral and cognitive symptoms in individuals with mental disorders.

Production and use of lentivirus to selectively transduce primary oligodendrocyte precursor cells for in vitro myelination assays.

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Myelination is a complex process that involves both neurons and the myelin forming glial cells, oligodendrocytes in the central nervous system (CNS) and Schwann cells in the peripheral nervous system (PNS). We use an in vitro myelination assay, an established model for studying CNS myelination in vitro. To do this, oligodendrocyte precursor cells (OPCs) are added to the purified primary rodent dorsal root ganglion (DRG) neurons to form myelinating co-cultures. In order to specifically interrogate the roles that particular proteins expressed by oligodendrocytes exert upon myelination we have developed protocols that selectively transduce OPCs using the lentivirus overexpressing wild type, constitutively active or dominant negative proteins before being seeded onto the DRG neurons. This allows us to specifically interrogate the roles of these oligodendroglial proteins in regulating myelination. The protocols can also be applied in the study of other cell types, thus providing an approach that allows selective manipulation of proteins expressed by a desired cell type, such as oligodendrocytes for the targeted study of signaling and compensation mechanisms. In conclusion, combining the in vitro myelination assay with lentiviral infected OPCs provides a strategic tool for the analysis of molecular mechanisms involved in myelination.
